



SR Research

SUPPORT

FAQ: How do I extract Reading Measures from Data Viewer? - Printable Version

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How do I extract Reading Measures from Data Viewer? - [Sam](#) - 09-14-2020

In reading research it is common to report standard "reading measures" which refer to a critical word or phrase in the text. Common reading measures are defined below.

FIXATION DURATION MEASURES

1. **Single Fixation Duration** - The duration of a fixation on a target region, provided that region is fixated one time only, and that the one fixation does not occur after fixations on words further along in the text.
2. **First Fixation Duration** - The duration of the first fixation on a target region, provided that the first fixation does not occur after fixations on words further along in the text.
3. **Gaze Duration** - The total duration of all fixations in a target region until the eyes fixate a region of text that is either progressive or regressive to the target region, provided that the first fixation on the target region does not occur after any fixations on words further along in the text.
 - Note: This measure is typically referred to as "Gaze Duration" for one-word regions of text and "First-Pass Duration" for multi-word regions of text. For simplicity in this reading guide, this measure will be referred to as " Gaze Duration" , whether the region is one word in length or more than one word in length.
4. **Right-Bounded Duration** - The total duration of all fixations in a target region until the eyes fixate a region of text that is progressive to the target region, provided that the first fixation on the target region does not occur after any fixations on words further along in the text.
 - Note: This measure, as well as Regression Path Duration below, is sometimes referred to as Go-Past Duration. These two definitions of Go-Past Duration are distinguished here by calling one Right-Bounded Duration and the other Regression Path Duration.
5. **Regression Path Duration** - The total duration of all fixations that occur from the first fixation on a target region until the target region is exited in a progressive manner (including fixations on the target region and fixations on words regressive to the target region), provided that the first fixation on the target region does not occur after any fixations on words further along in the text.
 - Note: This measure, as well as Right-Bounded Duration above, is sometimes referred to as Go-Past Duration. These two definitions of Go-Past Duration are distinguished here by calling one Right-Bounded Duration and the other Regression Path Duration.
6. **Re-reading Duration** - The difference between Regression Path Duration and Gaze Duration.
7. **Total Duration** - The total duration of all fixations in a target region.

FIXATION RATIO MEASURES

1. **First-Pass Regression Ratio** - The proportion of fixations following fixation on the target region that are regressive relative to the target region (and not progressive), provided that words: 1) further along in the text have not yet been fixated (i.e., the region was not skipped during first-pass reading) and that 2) the target region has not already been fixated and exited.
2. **First-Pass Fixation Ratio** - The proportion of times that the target region is fixated before any progressive text is fixated (as opposed to times when the target region is skipped during first-pass reading)
 - Note: First-Pass Fixation Ratio is First-Pass Skipping Ratio.

3. **First-Pass Multi-Fixation Ratio** - The proportion of times that the target region is fixated more than one consecutive time before any other text that is progressive or regressive to the region is fixated, provided that the target region is fixated at least once before any material progressive to the region is fixated (i.e., provided that the target region is not skipped during first-pass reading).
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Extracting Reading Measures from Data Viewer

There are two methods to extract Reading Measures from Data Viewer:

METHOD 1

The first approach, which provides a great deal of flexibility with respect to defining the critical word or words, is to use the Get Reading Measures tool, which can be downloaded from the thread below (which also contains detailed instructions for use):

[Get Reading Measures Application](#)

METHOD 2:

Alternatively, you can extract reading measures using a Data Viewer Interest Area Report:

To use this method the target regions (and any non-target regions) must be defined properly as Interest Areas in Data Viewer before creating the Interest Area Report. The advantage to this method as compared to using the Get Reading Measures tool is that the output from Data Viewer will contain the values of these dependent measures directly. The disadvantage of this method is that if you want to generate these dependent measures for regions of text that are not the same as the currently defined Interest Areas (e.g., if you want to get Gaze Durations on a region composed of words 4 to 6 and currently your viewing session has Interest Areas around each word individually) then you will need to redefine your Interest Areas before generating the Interest Area Report. You can do so by using the Merge option in Data Viewer (Edit -> Merge) OR you can do so by doing a dummy run of your experiment in Experiment Builder using different Automatic Interest Area Segmentation options (Edit -> Preferences -> Built-In Interest Area Preference -> WORD_SEGMENT) in order to generate new Interest Area Set files (IAS files). If you use the dummy run method you will then need to apply these new IAS files to your Data Viewer Viewing Session. This method of redefining Interest Areas is much more time-consuming than using the Get Reading Measures tool.

Reading measures can be extracted from a Data Viewer Interest Area Report as follows (you will need to make sure to include the variables listed in the definition of the measure in your Interest Area Report):

- **Single Fixation Duration** - IA_FIRST_FIXATION_DURATION, if IA_FIXATION_COUNT == 1 and IA_FIRST_FIX_PROGRESSIVE == 1
- **First Fixation Duration** - IA_FIRST_FIXATION_DURATION, if IA_FIRST_FIX_PROGRESSIVE == 1
- **Gaze Duration (a.k.a. First-Pass Duration)** - IA_FIRST_RUN_DWELL_TIME, if IA_FIRST_FIX_PROGRESSIVE == 1
- **Right-Bounded Duration (a.k.a. Go-Past Duration)** - IA_SELECTIVE_REGRESSION_PATH_DURATION, if IA_FIRST_FIX_PROGRESSIVE == 1
- **Regression Path Duration (a.k.a. Go-Past Duration)** - IA_REGRESSION_PATH_DURATION, if IA_FIRST_FIX_PROGRESSIVE == 1
- **Re-reading Duration** - [IA_REGRESSION_PATH_DURATION, if IA_FIRST_FIX_PROGRESSIVE == 1] - [IA_FIRST_RUN_DWELL_TIME, if IA_FIRST_FIX_PROGRESSIVE == 1]
- **Total Duration** - IA_DWELL_TIME

- **First-Pass Regression Ratio** - check the frequency of IA_REGRESSION_OUT, if IA_FIRST_FIX_PROGRESSIVE == 1
- **First-Pass Fixation Ratio** - check the proportion of IA_SKIP and subtract it from 1
- **First-Pass Multi-Fixation Ratio** - check the proportion of IA_FIRST_RUN_FIXATION_COUNT > 1, if IA_FIRST_FIX_PROGRESSIVE == 1

Note that there are some differences in how this approach and Get Reading Measures handle fixations that don't fall in any Interest Areas (fixations not on any words). By default the IA report approach will treat those fixations as fixations away from the target region (because the Run ends when there is a fixation anywhere outside the Interest Area). You can change this default behavior by going to the Preferences tab of the Data Viewer Inspector panel, and clicking on the Output / Analysis bullet. Then below you can check the box marked "Use Nearest Interest Area for Outlier Fixations". This will treat

fixations that are not on any Interest Area as if they are on the Nearest Interest Area. Alternatively, you can right-click on data in the Inspector and use Clean Function. Then you can delete fixations outside of Interest Areas. The Get Reading Measures tool effectively ignores fixations that are not in any Interest Area.
